EXCEL FUNCTIONS STUDY HANDBOOK

Lessons 88-105 | Nested IF, AI-assisted formulas, rounding, statistics and dynamic arrays

INCLUDED	WHAT YOU WILL PRACTICE
88-93 Nested logic	IF, AND, OR, complex grading, commission and performance policies
94-98 Rounding	ROUND, ROUNDUP, ROUNDDOWN, MROUND, CEILING and FLOOR
99-104 Statistical analysis	SMALL, LARGE, RANK, PERCENTILE, TREND, GROWTH and FREQUENCY
105 Dynamic arrays	UNIQUE, spill behavior, SORT and distinct-value counts

Each lesson contains copyable formulas, sample outputs and practical checks. Use the formulas in your own workbook, or reproduce the datasets shown in the lessons.

Scope: This handbook compiles the lessons covered in the current sequence (88-105). Earlier lookup-function lessons 77-87 are not reproduced here. The UNIQUE chapter is supplementary: the provided course transcript introduced UNIQUE but did not include its full demonstration.

Quick Reference: Which Function Should I Use?

Need	Excel function	Example
Choose among several conditions	IF / nested IF	=IF(B2>=60,"Pass","Fail")
Combine conditions	AND / OR	=AND(B2>=60,C2="Yes")
Nearest decimal / power of 10	ROUND	=ROUND(787.869,-1)
Away from / toward zero	ROUNDUP / ROUNDDOWN	=ROUNDUP(456.122,0)
Nearest arbitrary multiple	MROUND	=MROUND(457,50)
Next higher / lower multiple	CEILING / FLOOR	=CEILING(411,50)
k-th low / high value	SMALL / LARGE	=LARGE(B2:B11,3)
Rank a particular value	RANK.EQ	=RANK.EQ(B2,$B$2:$B$11,0)
Percentile threshold	PERCENTILE.INC	=PERCENTILE.INC(B2:B12,0.75)
Forecast straight-line values	TREND	=TREND(C2:C7,A2:A7,A8)
Forecast exponential values	GROWTH	=GROWTH(C2:C7,A2:A7,A8)
Frequency bins	FREQUENCY	=FREQUENCY(B2:B16,D2:D5)
Distinct values	UNIQUE	=UNIQUE(C2:C13)

LESSON 88 | Introduction to Nested Formulas

A nested formula places one Excel function inside another. Excel evaluates the inner expression to supply an argument for the outer function.

Core formula / syntax

```
=PRODUCT(SUM(B2:B5),SUM(C2:C5))
```

Examples and expected outputs

Task or input	Excel formula	Expected result
Sum then multiply	=PRODUCT(SUM(B2:B5),SUM(C2:C5))	Product of the two totals
Row-wise expenditure	=SUMPRODUCT(B2:B5,C2:C5)	Sum of each B-row x C-row

Try it in Excel

Enter quantities in B2:B5 and unit costs in C2:C5. Compare the product of column totals with the sum of individual quantity x price rows.

Remember

- SUMPRODUCT is generally the correct choice for total expenditure; multiplying the two column sums is usually not.
- Nested formulas can be built from the inside outward.

Self-check

If quantities are 2, 3 and prices 10, 20, what is SUMPRODUCT? Answer: 2x10 + 3x20 = 80

LESSON 89 | Nested IF: Coaching Recommendations

Assign coaching actions based on marks. Our explicit rule: above 80 = no coaching; above 60 through 80 = recommend; 60 or below = mandatory.

Core formula / syntax

```
=IF(C2>80,"No Coaching Required",IF(C2>60,"Recommend Coaching","Mandatory Coaching"))
```

Examples and expected outputs

Task or input	Excel formula	Expected result
Marks = 95	Outer IF: 95>80	No Coaching Required
Marks = 78	Inner IF: 78>60	Recommend Coaching
Marks = 60	Both tests FALSE	Mandatory Coaching
Marks = 80	First FALSE, second TRUE	Recommend Coaching

Try it in Excel

Place student marks in C2:C9, enter the formula in D2 and fill down. Test 59, 60, 61, 80 and 81 to verify boundaries.

Remember

- Each IF has a TRUE and FALSE branch.
- Conditions must be tested in a useful order.
- The course wording around exactly 60 was ambiguous; this practice rule deliberately uses >60.

LESSON 90 | Nested IF: Alternative Logic

The same decisions can be produced by checking the lowest marks first. This version tests <=60, then >80, then uses the remaining case for recommendations.

Core formula / syntax

```
=IF(C2<=60,"Mandatory Coaching",IF(C2>80,"No Coaching Required","Recommend Coaching"))
```

Examples and expected outputs

Task or input	Excel formula	Expected result
56	=IF(56<=60,"Mandatory",IF(56>80,"No","Recommend"))	Mandatory
78	=IF(78<=60,"Mandatory",IF(78>80,"No","Recommend"))	Recommend
95	=IF(95<=60,"Mandatory",IF(95>80,"No","Recommend"))	No
Missing FALSE branch	=IF(1>2,"Yes")	FALSE
Empty FALSE argument	=IF(1>2,"Yes",)	0

Try it in Excel

Compare this formula with the Lesson 89 formula across the same set of marks. The results should match at every boundary.

Remember

- Omitting the FALSE argument produces logical FALSE; leaving it empty after a comma can produce 0.
- The final ELSE case should be intentionally defined.

LESSON 91 | Complex Nested IF: Five Grades

Build a grading formula from highest threshold to lowest: >90 A, >80 B, >70 C, >60 D, otherwise E.

Core formula / syntax

```
=IF(C2>90,"A",IF(C2>80,"B",IF(C2>70,"C",IF(C2>60,"D","E"))))
```

Examples and expected outputs

Task or input	Excel formula	Expected result
95	First threshold >90	A
90	Not >90; >80	B
80	Not >80; >70	C
65	Not >70; >60	D
60	No threshold met	E
IFS alternative	=IFS(C2>90,"A",C2>80,"B",C2>70,"C",C2>60,"D",TRUE,"E")	Same grade

Try it in Excel

Put marks in column C. Start by writing one IF, then replace the FALSE case with a second IF and keep nesting until all grades are covered.

Remember

- Four IF functions need four closing parentheses.
- Check high thresholds first when using > conditions.
- Test exactly 60, 70, 80 and 90.

LESSON 92 | ChatGPT-Assisted Sales Commission Formulas

Specify columns, first data row and exact commission slabs in the prompt. Illustrative slabs: <1000 0%; 1000-4999 5%; 5000-9999 7%; >=10000 10%. These are practice slabs, not verified original course rates.

Core formula / syntax

```
=IF(B4<1000,0,IF(B4<5000,B4*5%,IF(B4<10000,B4*7%,B4*10%)))
```

Examples and expected outputs

Task or input	Excel formula	Expected result
January: B4=8000	January formula in D4	560
February: C4=12000	=IF(C4<1000,0,IF(C4<5000,C4*5%,IF(C4<10000,C4*7%,C4*10%)))	1200
January: B4=300	January formula	0
Rate only vs amount	=IF(B4<1000,0,IF(B4<5000,5%,IF(B4<10000,7%,10%)))	7% for 8000

Try it in Excel

Use row 3 for headers. B=January sales; C=February sales; D=January commission; E=February commission; data starts row 4. Ask AI to generate D4 and E4 formulas and check a few outputs manually.

Remember

- A generated reference to B3 would be wrong: B3 contains a header.
- Check whether the required output is a percentage or a monetary amount.
- The transcript contains a numerical inconsistency: 560/8000 is 7%, not 9%.

LESSON 93 | ChatGPT-Assisted Performance Reviews

Combine department and rating rules. Illustrative policy: Marketing/Operations <1 review, <3 12%, <4 17%, else 24%; HR <2 review, <4 10%, else 20%.

Core formula / syntax

```
=IF(OR(C4="Marketing",C4="Operations"),IF(B4<1,"Performance Review",IF(B4<3,"12% Hike",IF(B4<4,"17% Hike","24% Hike"))),IF(C4="HR",IF(B4<2,"Performance Review",IF(B4<4,"10% Hike","20% Hike")),"Unknown Department"))
```

Examples and expected outputs

Task or input	Excel formula	Expected result
Marketing, 2.5	Use Marketing/Operations branch	12% Hike
Operations, 3.5	Use Marketing/Operations branch	17% Hike
HR, 1.8	Use HR branch	Performance Review
HR, 4.5	Use HR branch	20% Hike
Finance, 4.5	Unknown department fallback	Unknown Department

Try it in Excel

Arrange B=rating, C=department and D=result, first record on row 4. Ask ChatGPT to distinguish each department and test boundary ratings.

Remember

- OR checks either named department. AND is useful for interval checks.
- Never silently assign HR policy to an unknown department.
- Text such as "12% Hike" is not a numeric percentage for calculations.

LESSON 94 | ROUND

ROUND uses standard Excel rounding: positive values at halfway round away from zero. num_digits=0 rounds to whole numbers; positive digits to decimals; negative digits to tens/hundreds.

Core formula / syntax

```
=ROUND(number,num_digits)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
7.85 to whole	=ROUND(7.85,0)	8
7.85 to one decimal	=ROUND(7.85,1)	7.9
77.869 to two decimals	=ROUND(77.869,2)	77.87
787.869 to nearest 10	=ROUND(787.869,-1)	790
787.869 to nearest 100	=ROUND(787.869,-2)	800

Try it in Excel

Enter values in B2:B11 and put =ROUND(B2,0), =ROUND(B2,1) and =ROUND(B2,-1) in adjacent columns; copy down.

Remember

- Changing number formatting affects display, not the stored numerical value.
- ROUND can increase or decrease a positive value.

LESSON 95 | ROUNDUP

ROUNDUP always rounds away from zero when digits must be removed. For positive values it goes upward; for negative values it goes more negative.

Core formula / syntax

```
=ROUNDUP(number,num_digits)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
7.12 to whole	=ROUNDUP(7.12,0)	8
456.122 to whole	=ROUNDUP(456.122,0)	457
456.122 to 2 decimals	=ROUNDUP(456.122,2)	456.13
411.35 to next 10	=ROUNDUP(411.35,-1)	420
-7.12 to whole	=ROUNDUP(-7.12,0)	-8

Try it in Excel

Compare =ROUNDUP(B2,0) and =ROUND(B2,0) for 7.12, 7.85, 456.122 and -7.12.

Remember

- Exact multiples stay unchanged.
- The direction is away from zero, not always numerically upward.

LESSON 96 | ROUNDDOWN

ROUNDDOWN truncates at a chosen place value by rounding toward zero. For positive prices, it selects lower amounts; for negative numbers it moves toward zero.

Core formula / syntax

```
=ROUNDDOWN(number,num_digits)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
9.95 to whole	=ROUNDDOWN(9.95,0)	9
77.869 to 2 decimals	=ROUNDDOWN(77.869,2)	77.86
787.869 to lower 10	=ROUNDDOWN(787.869,-1)	780
787.869 to lower 100	=ROUNDDOWN(787.869,-2)	700
-7.85 to whole	=ROUNDDOWN(-7.85,0)	-7

Try it in Excel

For a column of positive prices, compare ROUND, ROUNDUP and ROUNDDOWN using num_digits = 0, 1, -1 and -2.

Remember

- Toward zero is not the same as toward negative infinity.
- Use ROUNDDOWN when the required business rule really calls for truncation.

LESSON 97 | MROUND

MROUND selects the nearest specified multiple, not necessarily the nearest power of ten. It accepts decimal multiples and rounds exact halfway cases away from zero.

Core formula / syntax

```
=MROUND(number,multiple)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
787 to nearest 50	=MROUND(787,50)	800
457 to nearest 50	=MROUND(457,50)	450
411 to nearest 50	=MROUND(411,50)	400
787 to nearest 25	=MROUND(787,25)	775

Task or input	Excel formula	Expected result
7.6 to nearest 0.25	=MROUND(7.6,0.25)	7.5
Opposite signs	=MROUND(-17,5)	#NUM!

Try it in Excel

Place a multiple such as 50 in H2. Use =MROUND(B2,$H$2) and change H2 to 5, 25, 50 or 100.

Remember

- MROUND may round up or down.
- Number and nonzero multiple generally must share the same sign.

LESSON 98 | CEILING and FLOOR

For positive numbers, CEILING returns the next higher multiple and FLOOR the next lower multiple. Compare both against MROUND, which chooses the nearest.

Core formula / syntax

```
=CEILING(number,significance) / =FLOOR(number,significance)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
411 to higher 50	=CEILING(411,50)	450
411 to lower 25	=FLOOR(411,25)	400
426 to lower 25	=FLOOR(426,25)	425
457 to nearest 50	=MROUND(457,50)	450
457 to higher 50	=CEILING(457,50)	500
25 units in boxes of 12	=CEILING(25,12)/12	3 boxes

Try it in Excel

Create three columns for =MROUND(B2,50), =CEILING(B2,50) and =FLOOR(B2,50). Use 411, 457, 526, 690.

Remember

- An exact multiple stays unchanged.
- CEILING.MATH and FLOOR.MATH have extra options for negative numbers.
- These positive-number directions are not a complete description of negative-number behavior.

LESSON 99 | SMALL

SMALL returns the k-th smallest numerical value, where k=1 is minimum. Sorting the sheet is not required.

Core formula / syntax

```
=SMALL(array,k)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
Smallest	=SMALL(B2:B11,1)	3200
Second-smallest	=SMALL(B2:B11,2)	4200
Third-smallest	=SMALL(B2:B11,3)	6700
Third distinct-smallest	=SMALL(UNIQUE(B2:B11),3)	Depends on duplicates

Try it in Excel

Sample B2:B11 values: 12500, 8500, 15600, 4200, 9800, 18500, 6700, 11200, 3200, 14300. Ask for k = 1, 2, 3.

Remember

- Duplicates count separately.
- k=0 or k beyond the count produces #NUM!.
- SMALL(range,1) equals MIN(range).

LESSON 100 | LARGE

LARGE returns the k-th largest numerical value. k=1 is the maximum and k=2 is the second-highest, including ties as separate positions.

Core formula / syntax

```
=LARGE(array,k)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
Largest	=LARGE(B2:B11,1)	18500
Second-largest	=LARGE(B2:B11,2)	15600
Third-largest	=LARGE(B2:B11,3)	14300
Fourth-largest	=LARGE(B2:B11,4)	12500
Maximum equivalent	=MAX(B2:B11)	18500

Try it in Excel

Use the same ten sales figures as Lesson 99. Compare LARGE(range,3) with SMALL(range,3).

Remember

- LARGE returns the value; RANK returns a position.
- Duplicates are counted separately.
- Invalid k produces #NUM!.

LESSON 101 | RANK and Ties

RANK assigns the position of a number among other numbers. Use 0 (or omit order) for highest-first, 1 for lowest-first.

Core formula / syntax

```
=RANK(B2,$B$2:$B$11,0)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
Ravi: 12500 descending	=RANK(12500,{12500,18500,8500,15600,4200,9800,18500,6700,14300,3200},0)	5
Priya: 18500 descending	Same dataset, descending	1
Kiran: 18500 descending	Same dataset, descending	1
Sneha: 15600 descending	Same dataset, descending	3
Pooja: 3200 ascending	Same dataset, ascending	1

Try it in Excel

Use the ten employee sales figures from Lesson 101. Put =RANK(B2,$B$2:$B$11,0) in C2 and change the final argument to 1 in D2.

Remember

- Tied values receive equal ranks and subsequent positions are skipped.
- RANK.EQ is the modern equivalent; RANK.AVG averages occupied positions for ties.
- Absolute comparison ranges matter when filling down.

LESSON 102 | PERCENTILE

A percentile is a numerical threshold within a distribution. The traditional PERCENTILE function uses the inclusive positioning method; 0.5 gives the median.

Core formula / syntax

```
=PERCENTILE(array,k)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
25th percentile	=PERCENTILE(B2:B12,0.25)	35000
50th percentile	=PERCENTILE(B2:B12,0.5)	60000
75th percentile	=PERCENTILE(B2:B12,0.75)	85000
90th percentile	=PERCENTILE(B2:B12,0.9)	100000
75th exclusive	=PERCENTILE.EXC(B2:B12,0.75)	90000

Try it in Excel

Enter B2:B12 = 10000,20000,30000,40000,50000,60000,70000,80000,90000,100000,110000. Evaluate k=.25, .5 and .75.

Remember

- k for PERCENTILE/PERCENTILE.INC must be 0 through 1.
- Results may interpolate between original observations.
- Percentile is not the same as percent of total sales.

LESSON 103 | TREND and GROWTH

TREND fits a linear regression; GROWTH fits an exponential regression. Historical y-values and x-values must match in length. GROWTH requires positive y-values.

Core formula / syntax

```
=TREND(known_y,known_x,new_x) / =GROWTH(known_y,known_x,new_x)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
Linear: July, month 7	=TREND(C2:C7,A2:A7,A8)	22000
Linear: August, month 8	=TREND(C2:C7,A2:A7,A9)	24000
Exponential: July	=GROWTH(C2:C7,A2:A7,A8)	17715.61
Exponential: August	=GROWTH(C2:C7,A2:A7,A9)	19487.17

Try it in Excel

Use separate worksheets. Linear: x=1..6, y=10000,12000,...,20000. Exponential: x=1..6, y=10000,11000,12100,13310,14641,16105.1. Forecast months 7..9.

Remember

- Linear trends use roughly constant absolute changes; exponential trends use roughly constant percentage changes.
- Exponential decline is also possible if historical y-values remain positive.
- Extrapolation is not a guarantee of future performance.

LESSON 104 | FREQUENCY

FREQUENCY counts numeric observations in successive bins and returns one extra count for observations above the last bin. Upper boundaries are inclusive.

Core formula / syntax

```
=FREQUENCY(data_array,bins_array)
```

Examples and expected outputs

Task or input	Excel formula	Expected result
15 student marks; bins 50,100,150,200	=FREQUENCY(B2:B16,D2:D5)	3, 4, 4, 3, 1
Count <=50	=COUNTIF(B2:B16,"<=50")	3
Count >50 and <=100	=COUNTIFS(B2:B16,">50",B2:B16,"<=100")	4
Total numeric values	=COUNT(B2:B16)	15

Try it in Excel

Marks: 35,42,65,78,85,110,125,145,170,195,50,100,150,200,220 in B2:B16. Bins 50,100,150,200 in D2:D5. Enter formula in E2; modern Excel spills into E2:E6.

Remember

- Upper endpoints are included in their bin; the overflow bin is greater than the largest boundary.
- Spill results can be blocked (#SPILL!).
- Older Excel may require Ctrl+Shift+Enter for array formulas.

LESSON 105 | UNIQUE and Dynamic Arrays

UNIQUE extracts distinct values while leaving the source untouched. The optional exactly_once=TRUE asks for values that appear once in the source, which is different from the ordinary distinct list.

Core formula / syntax

```
=UNIQUE(array,[by_col],[exactly_once])
```

Examples and expected outputs

Task or input	Excel formula	Expected result
Distinct departments	=UNIQUE(C2:C13)	IT, HR, Finance, Sales, Marketing, Legal
Exactly once	=UNIQUE(C2:C13,FALSE,TRUE)	Legal
Sorted distinct	=SORT(UNIQUE(C2:C13))	Finance, HR, IT, Legal, Marketing, Sales

Task or input	Excel formula	Expected result
Count distinct	=COUNTA(UNIQUE(C2:C13))	6

Try it in Excel

Departments in C2:C13: IT, HR, IT, Finance, Sales, HR, IT, Marketing, Finance, Sales, Legal, Marketing. Place =UNIQUE(C2:C13) in E2 with empty cells below.

Remember

- UNIQUE is a dynamic-array function supported by recent Excel versions.
- A blocked result can produce #SPILL!.
- UNIQUE on multiple columns checks distinct row combinations.
- This lesson is supplemental to the transcript, which only began introducing UNIQUE.

Final Review: Boundary and Error Checklist

Function	What to test
Nested IF	All threshold boundaries; missing FALSE branch; quote marks; department fallback
Rounding	Positive and negative numbers; halfway cases; exact multiples
SMALL / LARGE	k=1, duplicates, k greater than available numeric observations
RANK	Order=0 versus 1; tied values; absolute ranges
PERCENTILE	k as a decimal from 0 to 1; interpolation; INC versus EXC
TREND / GROWTH	Matching known_x and known_y lengths; positive y-values for GROWTH
FREQUENCY	Bin boundaries inclusive; bins+1 outputs; empty spill cells
UNIQUE	Distinct versus exactly once; empty spill range; new source rows

Recommended practice sequence

Re-create one small worksheet for each lesson, verify at least three expected outputs, then change one boundary value or input to confirm you understand why the result changes.

End of the current lesson set: 88-105.